



2013
HIGHER SCHOOL CERTIFICATE
TRIAL EXAMINATION

Mathematics

Extension 2

General Instructions

- Reading Time - 5 minutes
- Working Time - 3 hours
- Write using a blue or black pen. Black pen is preferred
- Board approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- Show all necessary working in Questions 11-16

Total marks (100)

Section I

Total marks (10)

- Attempt Questions 1-10
- Answer on the Multiple Choice answer sheet provided
- Allow about 15 minutes for this section

Section II

Total marks (90)

- Attempt questions 11 – 16
- Answer on the blank paper provided, unless otherwise instructed
- Start a new page for each question
- All necessary working should be shown for every question
- Allow about 2 hours 45 minutes for this section

Section I**10 marks****Attempt Questions 1-10****Allow about 15 minutes for this section**

Use the multiple choice answer sheet for Questions 1 – 10.

1. If $Z_1 = 5 - 2i$ and $Z_2 = 3 + 4i$ then $Z_1 \overline{Z_2} =$

(A) $23 + 14i$

(B) $7 + 26i$

(C) $7 - 26i$

(D) $23 - 26i$

2. The equation of an ellipse is given by $4x^2 + 9y^2 = 36$. The foci and the directrices of this ellipse are:

(A) $(\pm\sqrt{5}, 0)$ and $x = \pm \frac{9\sqrt{5}}{5}$

(B) $(0, \pm\sqrt{5})$ and $x = \pm \frac{9\sqrt{5}}{5}$

(C) $(\pm\sqrt{5}, 0)$ and $y = \pm \frac{9\sqrt{5}}{5}$

(D) $(0, \pm\sqrt{5})$ and $y = \pm \frac{9\sqrt{5}}{5}$

3. If $\frac{x+1}{x^2-4} = \frac{a}{x+2} + \frac{b}{x-2}$ then:

(A) $a = -1$ $b = 3$

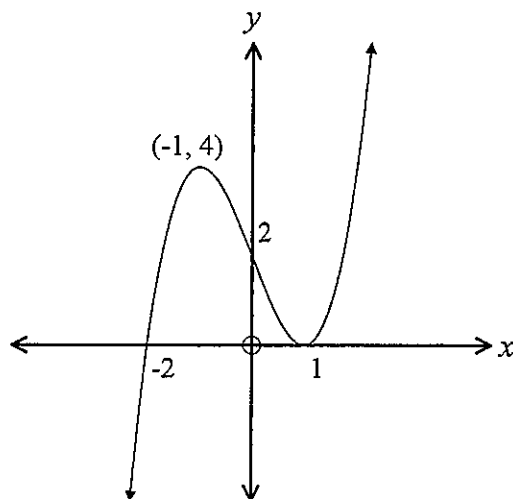
(B) $a = -\frac{1}{4}$ $b = \frac{3}{4}$

(C) $a = -\frac{1}{4}$ $b = \frac{1}{4}$

(D) $a = \frac{1}{4}$ $b = \frac{3}{4}$

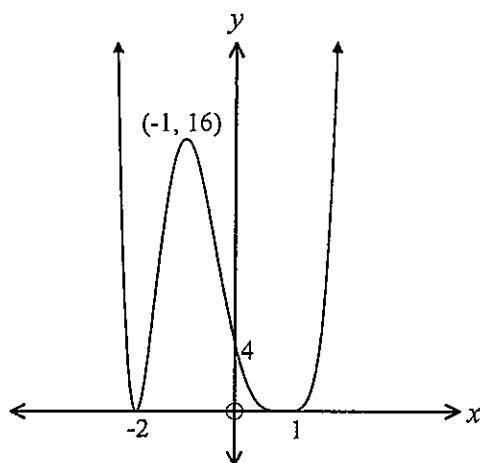
4. Given the curve $y = f(x)$, where $f(x)$ is defined for all real x , then the curve $y = -f(x)$ is best described by:
- (A) A reflection of $y = f(x)$ in the y -axis.
 - (B) A reflection of $y = f(x)$ in the x -axis.
 - (C) A reflection of $y = f(x)$ in the y -axis for $0 \leq x$.
 - (D) A reflection of $y = f(x)$ in the x -axis for $y \leq 0$.
5. The equation $x^3 + 2x^2 - 4x + 5 = 0$ has roots α , β and γ . The value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ is
- (A) $\frac{5}{4}$
 - (B) $-\frac{5}{4}$
 - (C) $\frac{4}{5}$
 - (D) $-\frac{4}{5}$
6. $\int \sin^3 x \, dx =$
- (A) $\cos^3 x - \cos x + c$
 - (B) $\frac{1}{3} \cos^3 x - \cos x + c$
 - (C) $\cos^3 x + \cos x + c$
 - (D) $\frac{1}{3} \cos^3 x + \cos x + c$

7. The graph of the function $y = f(x)$ is drawn below:

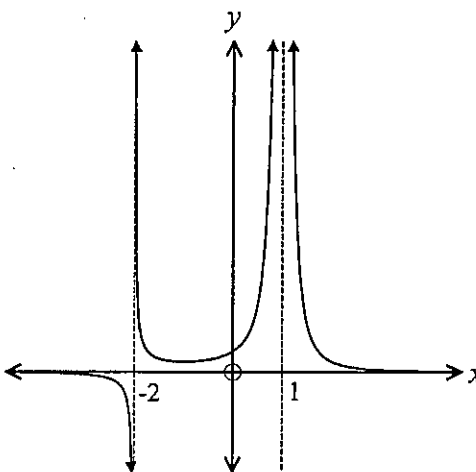


Which of the following graphs best represents the graph $y^2 = f(x)$

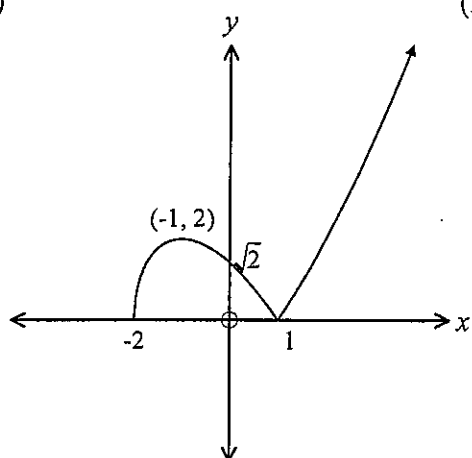
(A)



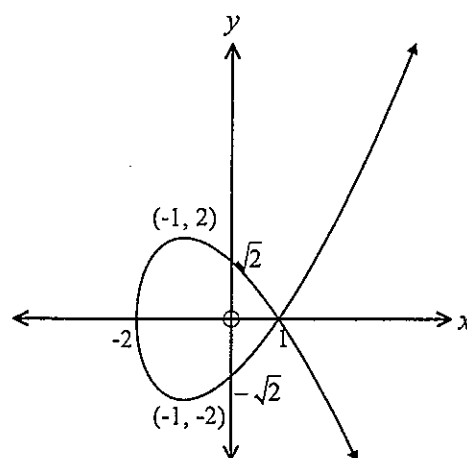
(B)



(C)



(D)



8. $\int_{-1}^1 \frac{1}{x^2 + 2x + 5} dx =$

(A) $\frac{\pi}{4}$

(B) $\frac{\pi}{8}$

(C) $\frac{\pi}{16}$

(D) 0

9. Given $3x + 2iy - ix + 5y = 7 + 5i$ where x and y are real numbers. Then

(A) $x = -1, y = 2$

(B) $x = \frac{39}{11}, y = -\frac{8}{11}$

(C) $x = -\frac{3}{5}, y = \frac{22}{5}$

(D) $x = -11, y = 8$

10. The solutions of the equation $z^3 = 8$ are $z =$

(A) $2, -\frac{1}{2} + \frac{i\sqrt{3}}{2}, -\frac{1}{2} - \frac{i\sqrt{3}}{2}$

(B) $2, \frac{1}{2} + \frac{i\sqrt{3}}{2}, \frac{1}{2} - \frac{i\sqrt{3}}{2}$

(C) $2, -1 + i\sqrt{3}, -1 - i\sqrt{3}$

(D) $2, 1 + i\sqrt{3}, 1 - i\sqrt{3}$

Section II**Total marks (90)****Attempt Questions 11-16****Allow about 2 hours 45 minutes for this section**

Answer all questions, starting each question in a new booklet with your name and question number on the front page. Do not write on the back of sheets.

Question 11 (15 marks)	Use a separate booklet	Marks
a)	i) Graph $y = f(x)$ where $f(x) = 2x - x^2$	2
	ii) Hence sketch.	
	$\alpha)$ $y = \frac{1}{f(x)}$	1
	$\beta)$ $y = (f(x))^2$	1
b)	i) Write $\sqrt{3} + i$ in modulus argument form	2
	ii) hence evaluate $(\sqrt{3} + i)^6$.	1
c)	Use the substitution $t = \tan \frac{x}{2}$ to evaluate $\int_0^{\frac{\pi}{2}} \frac{2}{5 + 3 \cos x} dx$.	3
d)	The area between the co-ordinates axes and the line $2x + 3y = 6$ is rotated about the line $y = 3$. By taking slices perpendicular to the axis of rotation find the volume formed.	3
e)	Represent on an Argand diagram the region for which the inequalities $ z - 3 - 3i < 5$ and $\frac{\pi}{4} \leq \arg z \leq \frac{3\pi}{4}$ are both satisfied.	2

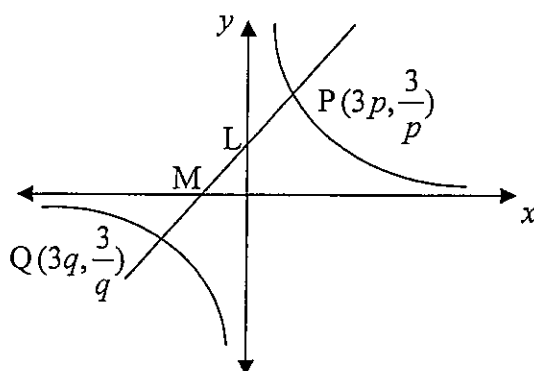
End of Question 11

Question 12 (15 marks)	Use a separate booklet	Marks
a) i) Find the Cartesian equation of the curve whose parametric equations are $x = 2 \cos \theta, y = \sin \theta$		2
ii) Find the equation of the normal to this curve at the point P where $\theta = \frac{\pi}{4}$		3
b) i) Show that the tangent from the origin to the curve $y = \ln x$ has a gradient of $\frac{1}{e}$		2
ii) Hence find the set of values of the real number k for which the equation $\ln x = kx$ has two distinct real roots.		1
c) i) Use the substitution $x = u^2$ ($u > 0$) to show that $\int_4^9 \frac{\sqrt{x}}{x-1} dx = 2 + \log_e \left(\frac{3}{2} \right).$		3
ii) Hence use integration by parts to find the value of $\int_4^9 \frac{1}{\sqrt{x}} \log_e (x-1) dx$		2
d) Describe the set of points in the complex plane that satisfies $ Z+1 = Z-i $		2

End of Question 12

Question 13 (15 marks) Use a separate booklet**Marks**

a)



A chord PQ of the rectangular hyperbola $xy = 9$ meets the asymptotes at L and M as shown.

- i) Show that the equation of chord PQ is: $pqy + x = 3(p + q)$ 2
 - ii) Find the coordinates of N, the mid-point of PQ 2
 - iii) Show that $PL = MQ$. 2
 - iv) If the chord PQ is a tangent to the parabola $y^2 = 3x$, find the locus of N 3
- b) If $1 - 2i$ is a root of the equation $z^2 - (3 + i)z + c = 0$.
- i) Explain why the conjugate $1 + 2i$ cannot be a root of the equation. 1
 - ii) Show that the other root is $2 + 3i$. 1
 - iii) Find the value of c . 1
- c) For a real number, r , the polynomial $P(x) = 8x^3 - 4x^2 - 42x + 45$ is divisible by $(x - r)^2$. Find r . 3

End of Question 13

Question 14 (15 marks) Use a separate booklet**Marks**

a)

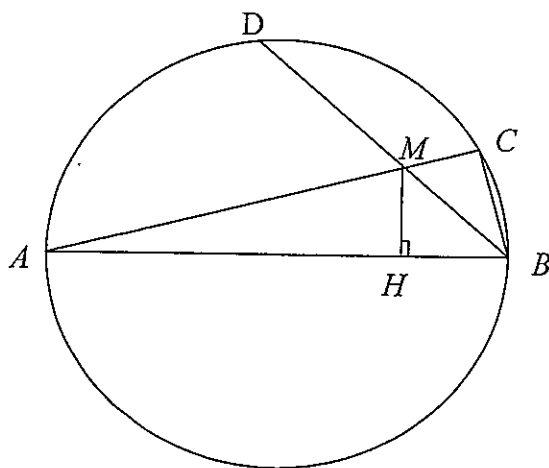
- i) Express $z = 2i$ and $w = -1 + \sqrt{3}i$ in modulus argument form. On an argand diagram plot the points P and Q which represent z and w .

2

- ii) On the same diagram construct vectors which represent $z + w$ and $z - w$. Hence deduce the exact values of $\arg(z + w)$ and $\arg(z - w)$.

2

b)



AB is the diameter of a circle. Chords AC and BD intersect at M. H is a point on AB such that MH is perpendicular to AB.

- i) Prove that triangle ABC is similar to triangle AMH.

2

- ii) Show that $AB \cdot AH = AC \cdot AM$.

1

- iii) Prove that $AB^2 = AC \cdot AM + BD \cdot BM$

2

- c) The area between the curve $y = 8x - x^2$, the x axis and the line $x = 4$ is rotated about the line $x = 4$. Find the volume generated by using:

- i) cylindrical shells.

3

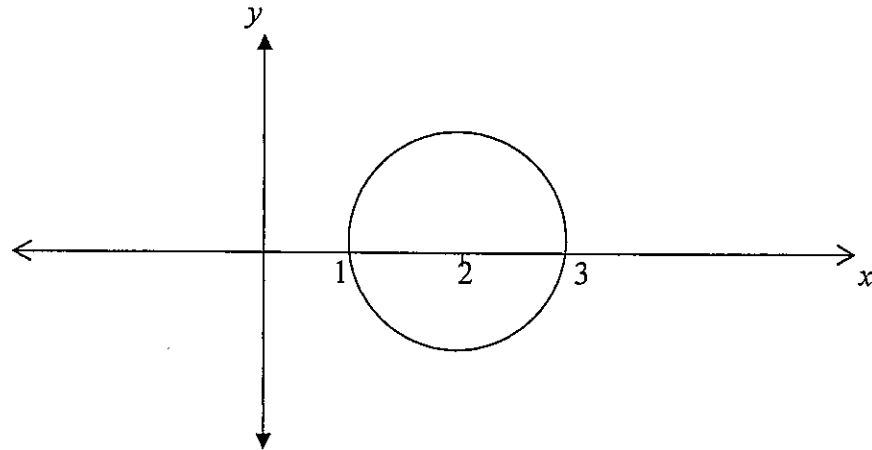
- ii) slicing.

3

End of Question 14

Question 15 (15 marks) Use a separate booklet**Marks**

a) i)



In the diagram above the circle $(x-2)^2 + y^2 = 1$ is drawn. The region bounded by the circle is rotated about the line $x = 1$. Use the method of cylindrical shells to show that the volume of the solid of revolution so formed is given by.

$$V = 4\pi \int_1^3 (x-1)\sqrt{1-(x-2)^2} dx \quad 2$$

ii) By using the substitution $x-2 = \sin u$, or otherwise, calculate the volume of the solid of revolution. 3

b) i) Find the three cube roots of unity. 2

ii) If ω is one of the complex roots of unity prove the other is ω^2 and show that $1 + \omega + \omega^2 = 0$. 2

iii) Prove that if n is a positive integer, then $1 + \omega^n + \omega^{2n} = 3$ or 0 depending on whether n is or is not a multiple of 3 . 3

c) The equation $x^3 + x^2 - 2x - 3 = 0$ has roots α, β and γ . Find the equation with roots α^2, β^2 and γ^2 3

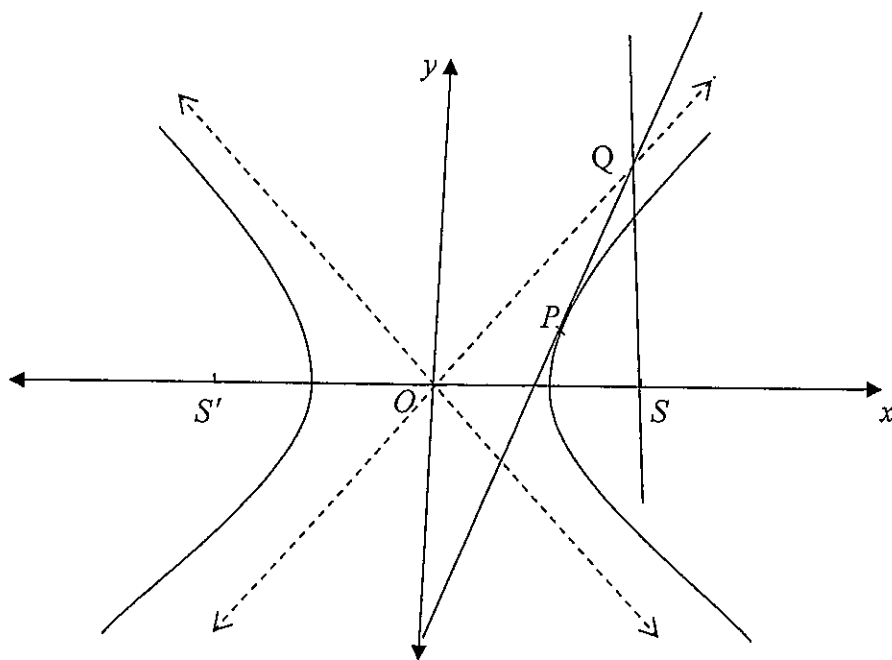
End of Question 15

Question 16 (15 marks) Use a separate booklet

a) Show that the derivative of $y = x^{x+1}$ is $\left(1 + \frac{1}{x} + \ln x\right)x^{x+1}$. 2

b) Sketch the graph of $y = \frac{2+x-x^2}{(x-1)^2}$ clearly showing any turning points and any asymptotes. 4

c)



d) The point $P(a \sec \theta, b \tan \theta)$ on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ with focus, S , is such that the tangent at P , the latus rectum through S , and one asymptote are concurrent. Prove that SP is parallel to the other asymptote. 4

(you may assume the equation of the tangent at P is $\frac{x \sec \theta}{a} - \frac{y \tan \theta}{b} = 1$)

e) i) Show that $(1 - \sqrt{x})^{n-1} \cdot \sqrt{x} = (1 - \sqrt{x})^{n-1} - (1 - \sqrt{x})^n$ 2

ii) If $I_n = \int_0^1 (1 - \sqrt{x})^n dx$ for $n \geq 0$ show that $I_n = \frac{n}{n+2} I_{n-1}$ for $n \geq 1$ 3

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

①

EXT 2 TRIAL SOLUTIONS 2013

$$1) (5-2i)(3-4i)$$

$$= 15 - 20i - 6i - 8$$

$$= 7 - 26i$$

(C)

$$2) 4x^2 + 9y^2 = 36$$

$$\frac{x^2}{9} + \frac{y^2}{4} = 1$$

$$b^2 = a^2(1 - e^2)$$

$$4 = 9(1 - e^2)$$

$$\frac{4}{9} = 1 - e^2$$

$$e^2 = \frac{5}{9}$$

$$e = \frac{\sqrt{5}}{3}$$

focus $(\pm ae, 0)$

$$= (\pm \sqrt{5}, 0)$$

directrix $x = \pm \frac{a}{e}$

$$= \pm \frac{9}{\sqrt{5}}$$

$$\therefore = \pm \frac{9\sqrt{5}}{5}$$

(A)

$$3) \frac{x+1}{x^2-4} = \frac{a}{x+2} + \frac{b}{x-2}$$

$$x+1 = a(x-2) + b(x+2)$$

$$\text{let } x=2: 3 = 4b$$

$$b = \frac{3}{4}$$

$$\text{let } x=-2: -1 = -4a$$

$$a = \frac{1}{4}$$

(D)

(4)

(B)

$$5) x^3 + 2x^2 - 4x + 5 = 0$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$= -4$$

$$\alpha\beta\gamma = -\frac{d}{a}$$

$$= -5$$

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$$

$$= \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma}$$

$$= \frac{-4}{-5}$$

$$= \frac{4}{5}$$

(C)

$$6) \int \sin^3 x \, dx$$

$$= \int \sin x \cdot \sin^2 x \, dx$$

$$= \int \sin x (1 - \cos^2 x) \, dx$$

$$= \int \sin x - \sin x \cos^2 x \, dx$$

$$= -\cos x + \frac{\cos^3 x}{3} + C$$

(B)

7) (D)

$$8) \int_{-1}^1 \frac{1}{x^2+2x+5} \, dx$$

$$= \int_{-1}^1 \frac{1}{x^2+2x+1+4} \, dx$$

$$= \int_{-1}^1 \frac{1}{(x+1)^2+4} \, dx$$

$$= \left[\frac{1}{2} \tan^{-1} \left(\frac{x+1}{2} \right) \right]_{-1}^1$$

$$= \frac{1}{2} (\tan^{-1} 1 - \tan^{-1} 0)$$

$$= \frac{1}{2} \left(\frac{\pi}{4} - 0 \right)$$

$$= \frac{\pi}{4}$$

(2)

$$9) 3x + 2iy - ix + 5y = 7 + 5i$$

$$\therefore 3x + 5y = 7 \quad \dots (1)$$

$$-x + 2y = 5 \quad \dots (2)$$

$$(2) \times 3 \quad -3x + 6y = 15 \quad \dots (3)$$

$$(1) + (3) \quad 11y = 22$$

$$y = 2$$

$$\text{Sub into (1)} \quad 3x + 10 = 7$$

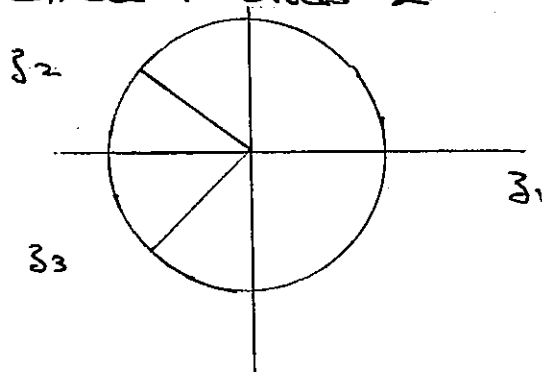
$$3x = -3$$

$$x = -1$$

$\therefore (A)$

$$10) z^3 = 8$$

zeros equally spaced around a circle radius 2.



$$z = 2$$

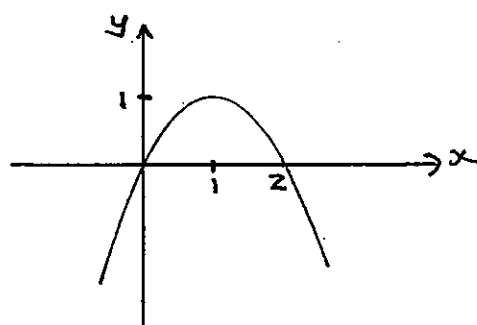
$$z_2 = 2 \operatorname{cis} \frac{2\pi}{3} = 2\left(-\frac{1}{2} + \frac{j\sqrt{3}}{2}\right) = -1 + j\sqrt{3}$$

$$z_3 = 2 \operatorname{cis} \left(-\frac{2\pi}{3}\right) = 2\left(-\frac{1}{2} - \frac{j\sqrt{3}}{2}\right) = -1 - j\sqrt{3}$$

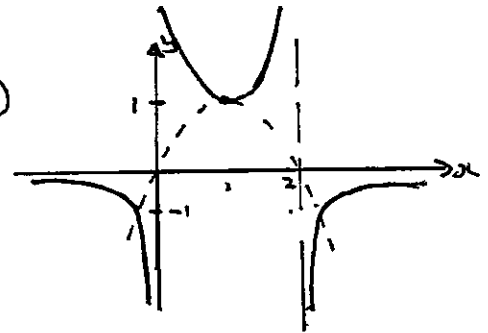
(2)

SECTION 11

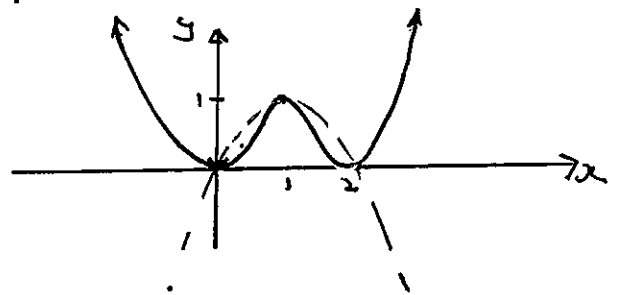
$$Q11) a) i) y = 2x - x^2$$



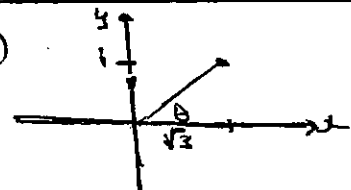
a)



b)



b) i)



$$\sqrt{3} + i = 2 \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right)$$

$$\begin{aligned} \text{(ii)} \quad (\sqrt{3} + i)^6 &= \left(2 \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right) \right)^6 \\ &= 64 (\cos \pi + i \sin \pi) \\ &= 64 (-1 + i0) \\ &= -64 \end{aligned}$$

$$c) \int_0^{\frac{\pi}{2}} \frac{2}{5 + 3 \cos x} dx.$$

$$t = \tan \frac{x}{2} : x=0, t=0 \\ x = \frac{\pi}{2}, t=1.$$

$$= \int_0^1 \frac{2}{5 + 3 \left(\frac{1-t^2}{1+t^2} \right)} \times \frac{2}{1+t^2} dt$$

$$= 4 \int_0^1 \frac{1}{5(1+t^2) + 3 - 3t^2} dt$$

$$= 4 \int_0^1 \frac{1}{2t^2 + 8} dt$$

(3)

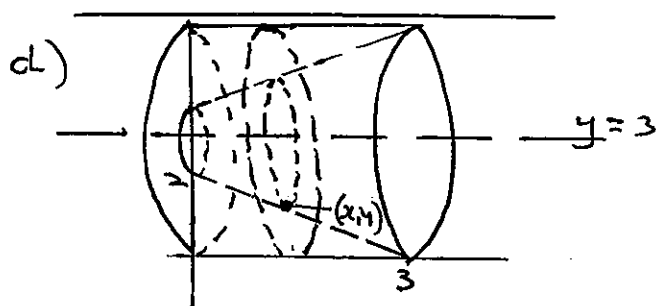
$$= 2 \int_0^1 \frac{1}{t^2+4} dt$$

$$= 2 \left[\frac{1}{2} \tan^{-1} \frac{t}{2} \right]$$

$$= \tan^{-1} \left(\frac{1}{2} \right) - \tan^{-1}(0)$$

$$= \tan^{-1} \left(\frac{1}{2} \right)$$

$$= 0.46 \text{ (2 dec. places)}$$



Volume of a Slice = $\pi (3^2 - (3-y)^2)$

$$V = \lim_{\Delta x \rightarrow 0} \sum_{x=0}^3 \pi (3^2 - (3-y)^2) \Delta x$$

$$V = \pi \int_0^3 3^2 - (3-y)^2 dx$$

$$= \pi \int_0^3 9 - (9 - 6y + y^2) dx$$

$$= \pi \int_0^3 6y - y^2 dx$$

$$= \pi \int_0^3 6 \left(\frac{6-2x}{3} \right) - \left(\frac{6-2x}{3} \right)^2 dx$$

$$= \pi \int_0^3 12 - 4x - \frac{1}{9} (6-2x)^2 dx$$

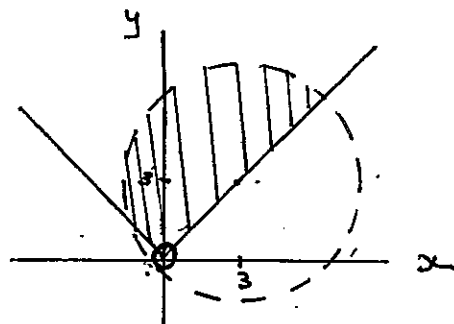
$$= \pi \left[12x - 2x^2 - \frac{1}{9} \left(\frac{6-2x}{-6} \right)^3 \right]_0^3$$

$$= \pi \left[12x - 2x^2 + \frac{(6-2x)^3}{54} \right]_0^3$$

$$= \pi [36 - 18 + 0 - (0 - 0 + 4)]$$

$$= 14\pi \text{ cubic units.}$$

e)



Q12)

a) i) $x = 2 \cos \theta \dots (1)$

$$y = \sin \theta \dots (2)$$

(1) $\Rightarrow \frac{x}{2} = \cos \theta \dots (3)$

$$(3)^2 + (2)^2 : \left(\frac{x}{2} \right)^2 + y^2 = \sin^2 \theta + \cos^2 \theta$$

$$\frac{x^2}{4} + y^2 = 1.$$

ii)

$$\frac{x^2}{4} + y^2 = 1$$

$$\frac{x}{2} + 2y \frac{dy}{dx} = 0$$

$$2y \frac{dy}{dx} = -\frac{x}{2}$$

$$\frac{dy}{dx} = -\frac{x}{4y}$$

$$\theta = \frac{\pi}{4} : x = 2 \cos \frac{\pi}{4} = \sqrt{2}$$

$$y = \sin \frac{\pi}{4} = \frac{\sqrt{2}}{2}$$

$$\therefore \frac{dy}{dx} = \frac{-\sqrt{2}}{2\sqrt{2}}$$

$$= -\frac{1}{2}$$

$$\therefore \text{gradient normal} = 2$$

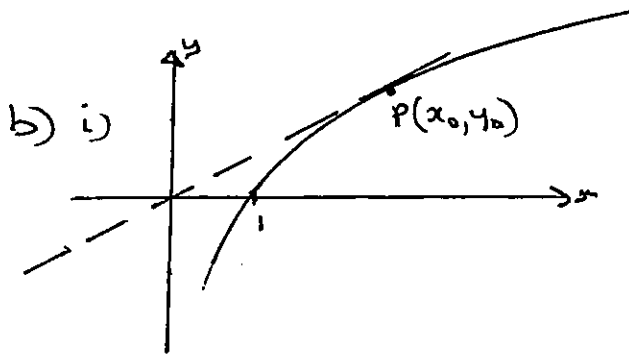
$$\therefore \text{eqn normal}$$

$$y - \frac{\sqrt{2}}{2} = 2(x - \sqrt{2})$$

$$y - \frac{\sqrt{2}}{2} = 2x - 2\sqrt{2}$$

$$2x - y - 2\sqrt{2} + \frac{\sqrt{2}}{2} = 0$$

(4)



let $P(x_0, y_0)$ be the point where the tangent hits the curve.

$$y = \ln x$$

$$\frac{dy}{dx} = \frac{1}{x}$$

at $x = x_0$ $\frac{dy}{dx} = \frac{1}{x_0}$

$\therefore$ eqn tangent: $y - y_0 = \frac{1}{x_0}(x - x_0)$

now this tangent passes through the origin:

$$\therefore 0 - y_0 = \frac{1}{x_0}(x - x_0)$$

$$-y_0 = -1$$

$$y_0 = 1$$

$\therefore (x_0, y_0)$ is on the curve

$$\therefore 1 = \log_e x_0$$

$$\therefore x_0 = e$$

Now gradient = $\frac{1}{x_0}$
 $= \frac{1}{e}$

ii) two distinct real roots

$\Rightarrow y = kx$ hits the curve $y = \ln x$ twice:

$$\therefore 0 < k < \frac{1}{e}$$

c) i) $\int_4^9 \frac{\sqrt{x}}{x-1} dx$

$$x = u^2 \quad \text{at } x = 4 \quad u = 2$$

$$\frac{dx}{du} = 2u \quad x = 9 \quad u = 3$$

$$dx = 2u \, du$$

$$\int_2^3 \frac{u}{u^2-1} \times 2u \, du$$

$$= 2 \int_2^3 \frac{u^2}{u^2-1} du$$

$$\frac{u^2}{u^2-1} = a + \frac{b}{u+1} + \frac{c}{u-1}$$

$$u^2 = a(u^2-1) + b(u-1) + c(u+1)$$

$$u=1: 1 = 2c$$

$$\frac{1}{2} = c$$

$$u=-1: 1 = -2b$$

$$-\frac{1}{2} = b$$

$$u=0: 0 = -a - b + c$$

$$0 = -a + \frac{1}{2} + \frac{1}{2}$$

$$a = 1$$

$$\therefore \frac{u^2}{u^2-1} = 1 - \frac{1}{2(u+1)} + \frac{1}{2(u-1)}$$

$$2 \int_2^3 \frac{u^2}{u^2-1}$$

$$\frac{2}{2} \int_2^3 2 - \frac{1}{u+1} + \frac{1}{u-1} du$$

$$= \left[2u - \ln(u+1) + \ln(u-1) \right]_2^3$$

$$= \left[2u + \ln\left(\frac{u-1}{u+1}\right) \right]_2^3$$

$$= \left(6 + \ln\left(\frac{2}{4}\right) \right) - \left(4 + \ln\left(\frac{1}{3}\right) \right)$$

$$= 6 - 4 + \ln\left(\frac{2}{1/3}\right)$$

$$= 2 + \ln\left(\frac{3}{2}\right)$$

ii) $\int_4^9 \frac{1}{\sqrt{x}} \ln(x-1) dx$

$$= \int_4^9 \frac{d}{dx} (2\sqrt{x}) \ln(x-1) dx$$

(5)

$$= \left[2\sqrt{x} \ln(x-1) \right]_4^9 - \int_4^9 \frac{2\sqrt{x}}{x-1} dx$$

$$= 6\ln 8 - 4\ln 3 - 2 \left(2 + \ln \left(\frac{3}{2} \right) \right)$$

$$= 6\ln 8 - 4\ln 3 - 4 - 2\ln \left(\frac{3}{2} \right)$$

$$= -4 + 2\ln \left(\frac{1024}{27} \right)$$

$$d) |z+1| = |z-i|$$

$$|x+iy+1| = |x+iy-i|$$

$$\sqrt{(x+1)^2 + y^2} = \sqrt{x^2 + (y-1)^2}$$

$$x^2 + 2x + 1 + y^2 = x^2 + y^2 - 2y + 1$$

$$2x = -2y$$

$$y = -x$$

$$Q13) i) m_{PQ} = \frac{\frac{3}{p} - \frac{3}{q}}{\frac{3p-3q}{3(q-p)}}$$

$$= \frac{\frac{3(q-p)}{pq}}{\frac{3(p-q)}{3(p-q)}}$$

$$= \frac{1}{-pq}$$

$$= -\frac{1}{pq}$$

$$\therefore \text{eqn } y - \frac{3}{p} = -\frac{1}{pq}(x - 3p)$$

$$pqy - 3q = -x + 3p$$

$$x + pqy = 3(p+q)$$

$$ii) N = \left(\frac{3p+3q}{2}, \frac{\frac{3}{p} + \frac{3}{q}}{2} \right)$$

$$= \left(\frac{3(p+q)}{2}, \frac{3}{2} \left(\frac{1}{p} + \frac{1}{q} \right) \right)$$

$$iii) L \left(0, \frac{3}{pq}(p+q) \right) M \left(3(p+q), 0 \right)$$

$$\text{mid pt LM} = \left(\frac{3(p+q)}{2}, \frac{3}{2pq}(p+q) \right)$$

$$= \left(\frac{3(p+q)}{2}, \frac{3}{2} \left(\frac{1}{p} + \frac{1}{q} \right) \right)$$

As this is the same mid pt as the mid pt of PQ then $PL = MQ$

$$iv) pqy + x = 3(p+q) \quad \text{--- (1)}$$

$$y^2 = 3x \quad \text{--- (2)}$$

$$(2) \Rightarrow x = \frac{y^2}{3}$$

Sub into (1)

$$pqy + \frac{y^2}{3} = 3(p+q)$$

$y^2 + 3pqy - 9(p+q) = 0$
as PQ is a tangent there must be only one solution
ie $\Delta = 0$

$$9p^2q^2 + 36(p+q) = 0$$

$$p^2q^2 = -4(p+q)$$

$$\therefore p+q = -\frac{p^2q^2}{4}$$

Now locus of N

$$x = \frac{3}{2}(p+q)$$

$$y = \frac{3}{2pq}(p+q)$$

$$\text{ie } x = \frac{3}{2} \times \frac{p^2q^2}{4}$$

$$x = -\frac{3p^2q^2}{8} \quad \text{--- (1)}$$

$$y = \frac{3}{2pq} \times \frac{-p^2q^2}{4}$$

$$y = -\frac{3pq}{8} \quad \text{--- (2)}$$

$$\text{from (2) } pq = -\frac{8y}{3}$$

Sub into (1)

$$x = -\frac{3}{8} \times \frac{64y^2}{9}$$

$$x = -\frac{8y^2}{3}$$

(6)

b) i) Conjugate is only a root if the coefficients are real. As coefficient of z is not real the conjugate cannot be a root.

ii) if α is the other root then $\alpha + 1 - 2i = -\frac{b}{a}$

$$\text{ie } \alpha + 1 - 2i = 3 + i$$

$$\alpha = 2 + 3i$$

$$\text{iii) } (1 - 2i)(2 + 3i) = \frac{c}{a}$$

$$(1 - 2i)(2 + 3i) = c$$

$$c = 2 + 3i - 4i + 6$$

$$= 8 - i$$

c) $8x^3 - 4x^2 - 42x + 45$
if divisible by $(x - r)^2$
the r is a double root.

$$P(x) = 8x^3 - 4x^2 - 42x + 45$$

$$P'(x) = 24x^2 - 8x - 42$$

$$P'(r) = 0$$

$$24r^2 - 8r - 42 = 0$$

$$12r^2 - 4r - 21 = 0$$

$$(6r + 7)(2r - 3) = 0$$

$$\therefore r = -\frac{7}{6}, \text{ or } \frac{3}{2}$$

$$\text{Now } P\left(\frac{3}{2}\right) = 8\left(\frac{27}{8}\right) - 4\left(\frac{9}{4}\right) - 63 + 45$$

$$= 27 - 9 - 63 + 45$$

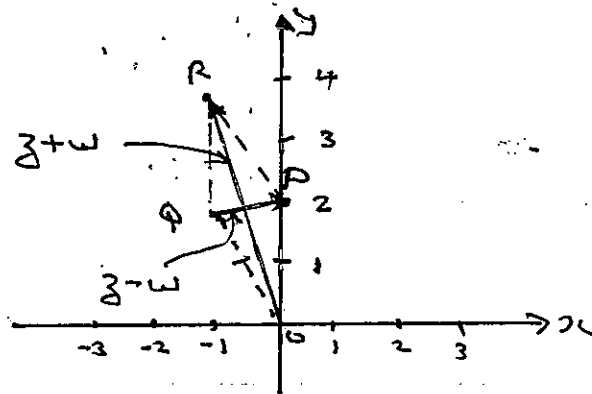
$$= 0$$

$$\therefore r = \frac{3}{2}$$

Q14) a) i)

$$z = 2i = 2\left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}\right)$$

$$w = -1 + \sqrt{3}i = 2\left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}\right)$$



As $OPRQ$ is a rhombus
($OP = OQ$)

$$\arg(z+w) = \frac{\pi}{2} + \left(\frac{1}{2}\left(\frac{2\pi}{3} - \frac{\pi}{2}\right)\right)$$

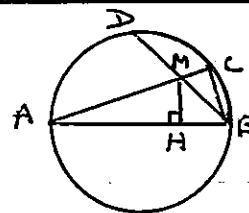
$$= \frac{\pi}{2} + \frac{\pi}{12}$$

$$= \frac{7\pi}{12}$$

$$\arg(z-w) = \frac{5\pi}{12} - \frac{\pi}{3}$$

$$= \frac{\pi}{12}$$

b)



i) $\triangle ABC \parallel \triangle AMH$

$\angle A$ is common

$\angle ACB = 90^\circ$ angle in a semi-circle - given AB diam

$\angle AHM = 90^\circ$ given

$\therefore \angle ACB = \angle AHM$

$\therefore \triangle ABC \parallel \triangle AMH$ equiangular

$$\text{ii) } \frac{AB}{AM} = \frac{AC}{AH}$$

$$AB \cdot AH = AM \cdot AC$$

iii) $\triangle BAD \parallel \triangle BMH$ (equiangular)
 $\angle B$ Common

$$\angle BHM = 90^\circ \text{ given}$$

$\angle BDA = 90^\circ$ angle in a semicircle

$$\therefore \frac{AB}{BM} = \frac{BD}{BH}$$

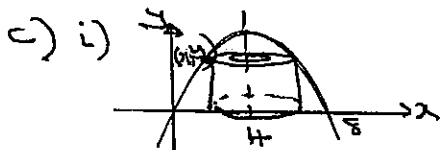
$$\therefore AB \cdot BH = BM \cdot BD$$

adding this to the result in (ii) gives

$$AB \cdot AH + AB \cdot BH = AM \cdot AC + BM \cdot BD$$

$$AB(AH + BH) = AM \cdot AC + BM \cdot BD$$

$$AB^2 = AC \cdot AM + BM \cdot BD$$



$$\text{Volume of a shell} = 2\pi(4-x)y \cdot \Delta x$$

$$V \approx \lim_{\Delta x \rightarrow 0} \sum_{x=0}^4 2\pi(4-x)(8x-x^2) \Delta x$$

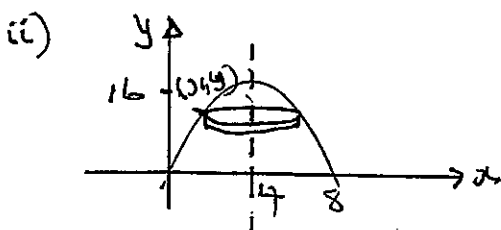
$$V = 2\pi \int_0^4 (4-x)(8x-x^2) dx$$

$$= 2\pi \int_0^4 (32x - 12x^2 + x^3) dx$$

$$= 2\pi \left[16x^2 - 4x^3 + \frac{x^4}{4} \right]_0^4$$

$$= 2\pi \left[(256 - 256 + \frac{256}{4}) - 0 \right]$$

$$= 128\pi \text{ cubic units}$$



$$\text{Volume of slice} = \pi(4-x)^2 \Delta y$$

$$V \approx \lim_{\Delta y \rightarrow 0} \sum_{y=0}^{16} \pi(4-x)^2 \Delta y$$

$$V = \pi \int_0^{16} (4-x)^2 dy$$

$$V = \pi \int_0^{16} (4 - (4 + \sqrt{16-y}))^2 dy$$

$$= \pi \int_0^{16} (16-y) dy$$

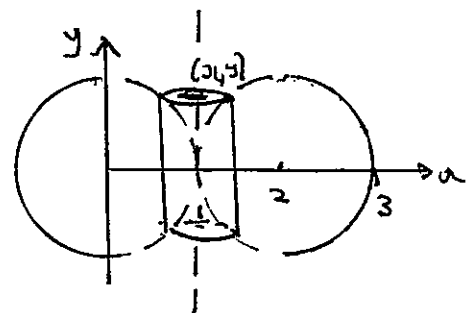
$$= \pi \left[16y - \frac{y^2}{2} \right]_0^{16}$$

$$= \pi (256 - 128 - 0)$$

$$= 128\pi \text{ cubic units}$$

Q15)

a) i)



$$\text{Volume of a shell} = 2\pi r h$$

$$= 2\pi(x-1)2y \Delta x$$

$$V \approx \lim_{\Delta x \rightarrow 0} 2\pi(x-1)2y \Delta x$$

$$V = 4\pi \int_1^3 (x-1)2y dx$$

$$= 4\pi \int_1^3 (x-1)\sqrt{1-(x-2)^2} dx$$

$$\left\{ \begin{array}{l} (x-2)^2 + y^2 = 1 \\ y^2 = 1 - (x-2)^2 \\ y = \sqrt{1 - (x-2)^2} \end{array} \right.$$

(18)

$$\text{ii) } V = 4\pi \int_1^3 (x-1) \sqrt{1-(x-2)^2} dx$$

$$\begin{aligned} x-2 &= \sin u & \begin{cases} x=1, & u = -\pi/2 \\ x=3, & u = \pi/2 \end{cases} \\ x &= \sin u + 2 \\ \frac{dx}{du} &= \cos u \\ dx &= \cos u du \end{aligned}$$

$$V = 4\pi \int_{-\pi/2}^{\pi/2} (\sin u + 1) \sqrt{1-\sin^2 u} \cos u du$$

$$= 4\pi \int_{-\pi/2}^{\pi/2} (\sin u + 1) \cos^2 u du$$

$$= 4\pi \int_{-\pi/2}^{\pi/2} \sin u \cos^2 u du + 4\pi \int_{-\pi/2}^{\pi/2} \cos^2 u du$$

$$= 4\pi \left[\frac{\cos^3 u}{3} \right]_{-\pi/2}^{\pi/2} + 2\pi \int_{-\pi/2}^{\pi/2} \cos 2u + 1 du$$

$$= 4\pi (0) + 2\pi \left[\frac{1}{2} \sin u + u \right]_{-\pi/2}^{\pi/2}$$

$$= 2\pi \left[\left(\frac{1}{2} \sin \pi + \pi \right) - \left(\frac{1}{2} \sin(-\pi) - \pi \right) \right]$$

$$= 2\pi \left[(0 + \pi) - (0 - \pi) \right]$$

$$= 2\pi (\pi) \text{ cubic units}$$

$$= 2\pi^2 \text{ units}^3$$

$$\text{b) i) } z^3 = 1$$

$$z^3 - 1 = 0$$

$$(z-1)(z^2+z+1) = 0$$

$$z=1 \quad z = \frac{-1 \pm \sqrt{1-4}}{2}$$

$$z = 1, \quad \frac{-1 + i\sqrt{3}}{2}, \quad \frac{-1 - i\sqrt{3}}{2}$$

$$\text{let } \omega = \frac{-1 + i\sqrt{3}}{2}$$

$$\therefore \omega^2 = \frac{1 - 2i\sqrt{3} - 3}{4}$$

$$= \frac{-2 - 2i\sqrt{3}}{4}$$

$$= \frac{-1 - i\sqrt{3}}{2}$$

= the other complex root.

$\therefore$ roots are $1, \omega, \omega^2$

Now $1 + \omega + \omega^2 = -\frac{b}{a}$ (sum of the roots)

$$= 0$$

iii) if n is a multiple of 3

let $n = 3k$

$$1 + \omega^{3k} + \omega^{6k}$$

$$= 1 + (\omega^3)^k + (\omega^3)^{2k}$$

$$= 1 + 1^k + 1^{2k}$$

$$= 1 + 1 + 1$$

$$= 3$$

if ' n ' is not a multiple of 3

let $n = 3k+1$ or $3k+2$

$$n = 3k+1: 1 + \omega^{3k+1} + \omega^{6k+2}$$

$$= 1 + \omega (\omega^3)^k + \omega^2 (\omega^3)^{2k}$$

$$= 1 + \omega + \omega^2$$

$$= 0$$

$$n = 3k+2: 1 + \omega^{3k+2} + \omega^{6k+4}$$

$$= 1 + \omega^2 (\omega^3)^k + \omega^4 (\omega^3)^{2k}$$

$$= 1 + \omega^2 + \omega^4$$

$$= 1 + \omega^2 + \omega^3 \cdot \omega$$

$$= 1 + \omega^2 + \omega$$

$$= 0$$

(9)

$$c) x^3 + x^2 - 2x - 3 = 0$$

Equation will be of the form.

$$(x^{\frac{1}{2}})^3 + (x^{\frac{1}{2}})^2 - 2(x^{\frac{1}{2}}) - 3 = 0$$

$$x^{\frac{3}{2}} + x - 2x^{\frac{1}{2}} - 3 = 0$$

$$x^{\frac{3}{2}} - 2x^{\frac{1}{2}} = 3 - x$$

$$x^{\frac{1}{2}}(x-2) = 3-x$$

$$x(x-2)^2 = (3-x)^2$$

$$x(x^2 - 4x + 4) = 9 - 6x + x^2$$

$$x^3 - 4x^2 + 4x = 9 - 6x + x^2$$

$$x^3 - 5x^2 + 10x - 9 = 0.$$

$$\Phi 16) a) y = x^{x+1}$$

$$\ln y = \ln x^{x+1}$$

$$\ln y = (x+1) \ln x$$

$$\ln y = x \ln x + \ln x$$

$$\frac{1}{y} \frac{dy}{dx} = x + \frac{1}{x} + \ln x + \frac{1}{x}$$

$$\frac{1}{y} \frac{dy}{dx} = 1 + \ln x + \frac{2}{x}$$

$$\frac{dy}{dx} = (1 + \ln x + \frac{2}{x}) y$$

$$= (1 + \frac{1}{x} + \ln x) x^{x+1}$$

$$b) y = \frac{2+x-x^2}{(x-1)^2}$$

asymptotes: $x = 1$

$$\text{horizontal: } \lim_{x \rightarrow \infty} \frac{\frac{2}{x^2} + \frac{x}{x^2} - \frac{x^2}{x^2}}{\frac{x^2}{x^2} - \frac{2x}{x^2} - \frac{1}{x^2}}$$

$$= \frac{0+0-1}{1-0-0}$$

$$\text{ie } y = -1.$$

'y' intercept: 2.

$$\text{'x' intercept: } 2+x-x^2 = 0$$

$$(2-x)(1+x) = 0$$

$$x = 2, -1.$$

$$\frac{dy}{dx} = \frac{(x-1)^2(1-2x) - 2(2+x-x^2)(x-1)}{(x-1)^4}$$

$$= \frac{(x-1)[(x-1)(1-2x) - 2(2+x-x^2)]}{(x-1)^4}$$

$$= \frac{(x-1)(x-5)}{(x-1)^4}$$

$$= \frac{x-5}{(x-1)^3}.$$

turning pts. $\frac{dy}{dx} = 0.$

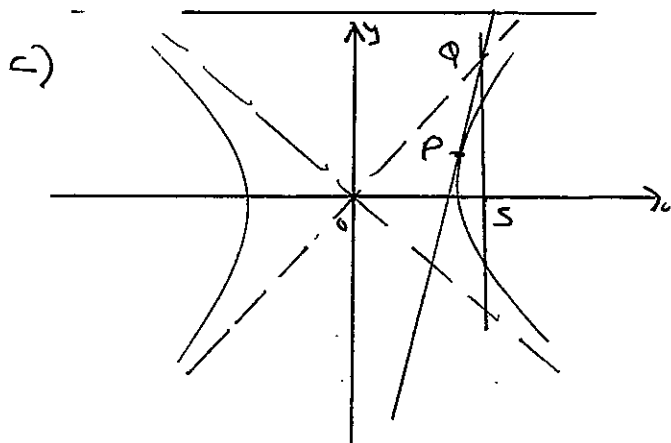
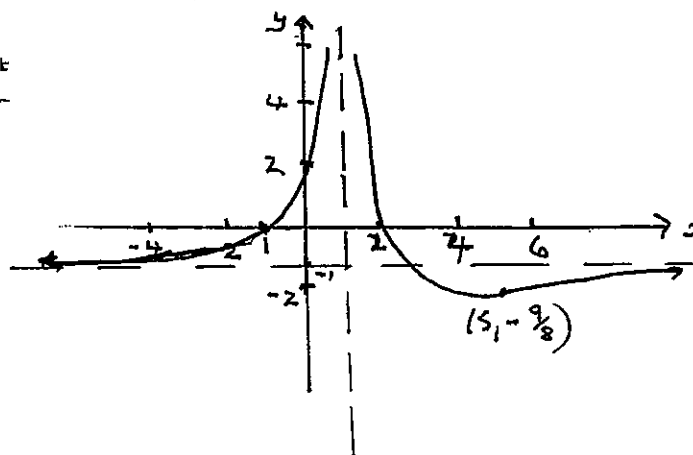
$$x-5 = 0$$

$$x = 5.$$

$$y = -\frac{9}{8}$$

$f(x)$	4	5	6
$f'(x)$	-	0	+

$\therefore (5, -\frac{9}{8})$ min.



(10)

eqn tangent at P: $\frac{x \sec \theta}{a} - \frac{y \tan \theta}{b} = 1$ (1) e) i) $(1 - \sqrt{x})^{n-1} \sqrt{x} = (1 - \sqrt{x})^{n-1} - (1 - \sqrt{x})^n$

eqn asymptote: $y = \frac{bx}{a}$... (2)

eqn latus rectum: $x = a^2$... (3)

Solving (2) and (3)

$Q(a^2, be)$

Sub into the equation of the tangent.

$$\frac{ae \sec \theta}{a} - \frac{be \tan \theta}{b} = 1$$

$$e \sec \theta - e \tan \theta = 1.$$

$$e = \frac{1}{\sec \theta - \tan \theta} \quad \dots (A)$$

Now gradient SP

$$= \frac{b \tan \theta}{a \sec \theta - a e}$$

$$= \frac{b \tan \theta}{a \sec \theta - a \left(\frac{1}{\sec \theta - \tan \theta} \right)} \quad \text{from (A)}$$

$$= \frac{b \tan \theta (\sec \theta - \tan \theta)}{a \sec \theta (\sec \theta - \tan \theta) - a}$$

$$= \frac{b \tan \theta \sec \theta - b \tan^2 \theta}{a \sec^2 \theta - a \sec \theta \tan \theta - a}$$

$$= \frac{b \tan \theta \sec \theta - b (\sec^2 \theta - 1)}{a (\sec^2 \theta - \sec \theta \tan \theta - 1)}$$

$$= \frac{b (\tan \theta \sec \theta - \sec^2 \theta + 1)}{a (\sec^2 \theta - \sec \theta \tan \theta - 1)}$$

$$= -\frac{b}{a}$$

which is the gradient of the other asymptote.

$\therefore SP \parallel$ to the other asymptote.

$$\begin{aligned} \text{RHS} &= (1 - \sqrt{x})^{n-1} - (1 - \sqrt{x})^n \\ &= (1 - \sqrt{x})^{n-1} (1 - (1 - \sqrt{x})) \\ &= (1 - \sqrt{x})^{n-1} (1 - 1 + \sqrt{x}) \\ &= (1 - \sqrt{x})^{n-1} \cdot \sqrt{x} \\ &= \text{L.H.S} \end{aligned}$$

$$\text{ii) } \int_0^1 (1 - \sqrt{x})^n$$

$$I_n = \int_0^1 \frac{d}{dx}(x) (1 - \sqrt{x})^n$$

$$= \left[x (1 - \sqrt{x})^n \right]_0^1 - n \int_0^1 x (1 - \sqrt{x})^{n-1} \cdot \frac{1}{2\sqrt{x}} dx$$

$$= 0 + \frac{n}{2} \int_0^1 \sqrt{x} (1 - \sqrt{x})^{n-1} dx$$

$$= \frac{n}{2} \int_0^1 (1 - \sqrt{x})^{n-1} dx - \frac{n}{2} \int_0^1 (1 - \sqrt{x})^n$$

$$I_n = \frac{n}{2} I_{n-1} - \frac{n}{2} I_n$$

$$2I_n = n I_{n-1} - n I_n$$

$$n I_n + 2I_n = n I_{n-1}$$

$$I_n (n+2) = n I_{n-1}$$

$$I_n = \frac{n}{n+2} I_{n-1}$$